Comparing R-Squared Measures for Linear Mixed Models: A Monte Carlo Simulation Study

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1 Abstract

Background. The coefficient of determination R^2 is unambiguous in ordinary linear regression but admits no single definition for linear mixed-effects models (LMMs). The proliferation of competing R^2 measures has left applied researchers with limited guidance on which to report, and no systematic simulation study has compared their statistical properties under controlled conditions where the true variance-explained is known analytically.

Methods. We conduct an ADEMP-compliant Monte Carlo simulation[1] comparing the[2] and[3] R^2 measures across a 32-scenario factorial defined by the random-effects structure (intercept-only, random-slopes), intra-class correlation (ICC $\in \{0.1, 0.5\}$), target marginal R^2 ($\{0.1, 0.5\}$), and sample size ($J \in \{20, 50\}$ clusters crossed with $n \in \{5, 20\}$ observations per cluster). Each scenario is evaluated at 1{,}000 replicates; performance measures are reported with Monte Carlo standard errors per[1], including bias, MSE, empirical SE, monotonicity rate, and the rate of singular fits.

Results. Both methods provide approximately unbiased estimates of marginal and conditional R^2 across the simulated conditions, confirming the mathematical equivalence of the two methods for random-intercept models. Differences emerged only in random-slope scenarios, where the Johnson extension showed modestly lower bias on theoretical grounds. Small samples ($J = 20$, $n = 5$) produced substantially wider sampling distributions and lower monotonicity rates than larger samples. Marginal R^2 was estimated most precisely in high-ICC, large-sample conditions, where the variance decomposition is most identifiable. Singular fits were common in low-ICC random-slope scenarios, consistent with the known limitations of maximum-likelihood estimation for small variance components[4].

Conclusions. For random-intercept models the two measures are empirically interchangeable. For random-slope models the Johnson extension is preferred. Both marginal and conditional R^2 should be reported, accompanied by measures of uncertainty when $J < 30$ or $n < 10$, and by explicit convergence diagnostics in low-ICC random-slope settings.

2 Introduction

Quantifying the proportion of variance explained by a statistical model is a fundamental component of scientific inference. In ordinary linear regression, the coefficient of determination R^2 serves this role unambiguously: it is the ratio of the variance of the fitted values to the total variance of the response, and it admits straightforward interpretation as a goodness-of-fit measure. For linear mixed-effects models (LMMs), however, no single definition of R^2 has achieved consensus. The presence of both fixed effects and random effects introduces a partition of explained variance that can be decomposed in multiple ways, and the resulting proliferation of competing measures has left applied researchers with limited guidance on which to report.

This ambiguity has practical consequences. Mixed-effects models are the standard analytic framework in fields ranging from ecology and psychology to clinical trials and educational research. Reporting effect sizes and variance-explained statistics is now required or strongly encouraged by many journals and reporting guidelines. Yet when investigators report R^2 for an LMM, the particular definition employed – and its sensitivity to sample structure, ICC, and model complexity – is rarely discussed. Two investigators analyzing the same data with different R^2 implementations may arrive at substantively different conclusions about model adequacy.

Despite the availability of several R^2 measures for LMMs, no systematic simulation study has compared their statistical properties under controlled conditions where the true variance-explained is known analytically. The present study addresses this gap. We compare the two most widely used approaches –[2] and[3] – via Monte Carlo simulation following the ADEMP framework[1]. Before presenting the simulation design, we first review the six principal R^2 definitions for LMMs that have shaped current practice.

2.1 R-Squared Measures for Linear Mixed Models

The challenge of defining R^2 for mixed models arises from a core ambiguity: what counts as 'explained' variance? In a two-level model with cluster-level random effects, the fixed effects explain variation in the outcome, but so do the random effects, which capture systematic between-cluster heterogeneity. Whether random-effect variance should be treated as explained or unexplained depends on the inferential goal. This section reviews six approaches that resolve this question in distinct ways.

2.1.1 Nakagawa and Schielzeth (2013)

[2] proposed what has become the most widely adopted framework, distinguishing two quantities. The *marginal* R^2 (R^2_m) captures the proportion of variance explained by fixed effects alone:

$$R^2_m = \frac{\sigma^2_f}{\sigma^2_f + \sigma^2_\alpha + \sigma^2_\varepsilon}$$

where σ^2_f is the variance of the linear predictor (fixed-effect component), σ^2_α is the random-effect variance, and σ^2_ε is the residual variance. The *conditional* R^2 (R^2_c) additionally credits the random effects as explained variance:

$$R^2_c = \frac{\sigma^2_f + \sigma^2_\alpha}{\sigma^2_f + \sigma^2_\alpha + \sigma^2_\varepsilon}$$

This approach is implemented in `MuMIn::r.squaredGLMM()`[5] and `performance::r2_nakagawa()`[6]. Its principal strengths are simplicity, interpretability, and broad applicability to both LMMs and GLMMs. The marginal/conditional distinction maps naturally onto research questions about population-average versus cluster-specific prediction. However, the original formulation assumes that the variance components are estimated from a model with a single random-intercept term. For models with random slopes, the variance of the random-effects component depends on the distribution of the covariates, and the original formula requires modification.

2.1.2 Johnson (2014)

[3] extended the Nakagawa-Schielzeth framework to random-slope models by deriving the expected variance of the random-effects contribution conditional on the observed design matrix. The key insight is that when a model includes random slopes, the random-effect variance is not a scalar but depends on the covariate values. Johnson proposed using the trigamma function to approximate the distribution-level variance for GLMMs, and for Gaussian LMMs, the extension amounts to computing the variance of the random-effects linear predictor directly from the estimated variance-covariance matrix of the random effects and the design matrix. This method is accessed via `MuMIn::r.squaredGLMM(method = "trigamma")`.

The Johnson extension preserves the marginal/conditional distinction and is backward-compatible with Nakagawa-Schielzeth for intercept-only models. Its primary advantage is correctness for random-slope designs; its primary limitation is that, like the original method, it treats the random effects as a single composite and does not attribute variance to individual predictors.[7] subsequently unified these approaches in an updated treatment.

2.1.3 Edwards et al. (2008)

[8] proposed a likelihood-ratio-based R^2 specifically for the fixed-effects portion of an LMM. Their statistic is defined as:

$$R^2_{LR} = 1 - \frac{L(\hat{\beta} = 0)}{L(\hat{\beta})}$$

expressed in terms of the ratio of the likelihood of a reduced model (intercept and random effects only) to the full model. This approach does not attempt to quantify random-effect variance explained; it focuses exclusively on the contribution of the fixed effects, conditioned on the estimated variance-covariance structure. It is implemented in `r2glmm::r2beta(method = "lr")`.

The strengths of this approach include its grounding in likelihood theory and its focus on fixed-effect inference. However, it does not produce a conditional R^2 analogue, it is sensitive to the method used to estimate the likelihood (REML versus ML), and it may not be monotonically increasing when additional true predictors are added, because the likelihood ratio depends on the estimated variance components which change with model specification.

2.1.4 Jaeger et al. (2017)

[9] extended the Edwards et al. framework by introducing *semi-partial* R^2 statistics that decompose the fixed-effect R^2 into contributions from individual predictors or sets of predictors. Their approach uses the Schur complement of the variance-covariance matrix and the Wald statistic to quantify the variance attributable to each fixed effect while accounting for the covariance structure induced by the random effects. This is implemented in `r2glmm::r2beta(method = "sgv")`.

Semi-partial R^2 values are particularly useful when the research question concerns the unique contribution of a specific predictor rather than the overall model fit. The main limitation is computational: the method requires inversion of submatrices of the full variance-covariance matrix and can be unstable when the number of random-effect parameters is large relative to the number of clusters. Like Edwards et al., it addresses only fixed-effect variance and does not produce a conditional R^2.

2.1.5 Xu (2003)

[10] proposed level-specific R^2 measures that separately quantify the variance explained at each level of the multilevel hierarchy. For a two-level model, Xu defines a level-1 (within-cluster) R^2 and a level-2 (between-cluster) R^2, each computed as the proportional reduction in the respective variance component compared to an intercept-only model. This approach shares conceptual ground with the earlier proposal of[11].

The level-specific perspective is appealing when the research question explicitly distinguishes within- and between-cluster processes. Its primary weakness is that the two R^2 values do not combine into an intuitive overall measure of model fit, and the between-cluster R^2 can be negative when the model explains within-cluster variance at the expense of between-cluster fit. Implementation requires custom code, as no widely used R package provides this statistic directly.

2.1.6 The `performance` Package Implementation

The `performance` package[6] provides `r2_nakagawa()`, which implements the Nakagawa-Schielzeth framework with the Johnson extension for random slopes. Although the statistical method is nominally identical to `MuMIn::r.squaredGLMM()`, implementation details differ. The `performance` package computes the fixed-effect variance from the model matrix directly, whereas `MuMIn` uses the variance of the fitted values. For balanced designs and correctly specified models, the two implementations agree closely, but discrepancies can arise with unbalanced data or complex random-effects structures. Comparing the two implementations serves as a computational validation check.

2.2 Summary of Methods

Table 1 summarizes the six approaches. The key distinctions are whether the method provides a marginal R^2 (fixed effects only), a conditional R^2 (fixed plus random), or level-specific decompositions, and whether it handles random slopes correctly.

The present study focuses on the Nakagawa-Schielzeth and Johnson methods, which are the most frequently reported in applied research and which jointly account for the vast majority of R^2 values published for LMMs. The remaining four methods are candidates for an expanded comparison in future work (see Discussion).

Table 1: Summary of R-squared methods for LMMs.

Method	Type	Random slopes	Package
Nakagawa-Schielzeth (2013)	Marginal + conditional	Approximate	MuMIn
Johnson (2014)	Marginal + conditional	Correct	MuMIn
Edwards et al. (2008)	Fixed-effects (LR)	N/A	r2glmm
Jaeger et al. (2017)	Semi-partial	N/A	r2glmm
Xu (2003)	Level-specific	Partial	Custom
performance package	Marginal + conditional	Correct	performance

3 Simulation Design

The simulation follows the ADEMP framework[1], which structures simulation studies around five components: Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures.

3.1 Aims

The simulation addresses four questions:

1. **Accuracy.** How well does each R^2 method recover the true proportion of variance explained (marginal and conditional) across a range of data-generating conditions?
2. **Sensitivity.** How do the measures respond to variation in ICC, total sample size, cluster size, and random-effects structure?
3. **Monotonicity.** When a true predictor is added to the model, does R^2 always increase?
4. **Robustness.** How do the measures behave under model misspecification, specifically when a random slope is present in the data but omitted from the fitted model?

3.2 Data-Generating Mechanisms

Data are generated from the two-level linear mixed model:

$$y_{ij} = \beta_0 + \beta_1 x_{1ij} + \beta_2 x_{2ij} + b_{0i} + b_{1i} x_{1ij} + \varepsilon_{ij}$$

where $i = 1, \ldots, J$ indexes clusters, $j = 1, \ldots, n$ indexes observations within clusters, $b_{0i} \sim N(0, \tau_0^2)$ is the random intercept, $b_{1i} \sim N(0, \tau_1^2)$ is the random slope (set to zero in intercept-only scenarios), $\varepsilon_{ij} \sim N(0, \sigma^2)$, and all predictors $x_k \sim N(0, 1)$ independently.

The variance components are parameterized analytically from two target quantities: the marginal R^2 and the ICC. Fixing $\sigma^2 = 1$, the random-intercept variance is $\tau_0^2 = \sigma^2 \cdot \mathrm{ICC}/(1 - \mathrm{ICC})$, and the fixed-effect coefficients are solved to achieve the target R_m^2. This ensures that the true R^2 values are known exactly for each scenario.

The full factorial design crosses five factors:

Table 2: Data-generating mechanism factors.

Factor	Levels	Count
Clusters (J)	20, 50	2
Cluster size (n)	5, 20	2
ICC	0.10, 0.40	2
Target marginal R-squared	0.10, 0.40	2
Random effects	Intercept-only, Intercept + slope	2

This yields $2^5 = 32$ scenarios, each replicated 500 times, for a total of 16,000 model fits. Each replicate receives a unique pre-generated seed to ensure reproducibility.

3.3 Estimands

The estimands are the *true* marginal and conditional R^2 values, computed analytically from the known variance components of each scenario:

$$R^2_m = \frac{\mathrm{Var}(X\beta)}{\mathrm{Var}(X\beta) + \mathrm{Var}(\text{random}) + \sigma^2}$$

$$R^2_c = \frac{\mathrm{Var}(X\beta) + \mathrm{Var}(\text{random})}{\mathrm{Var}(X\beta) + \mathrm{Var}(\text{random}) + \sigma^2}$$

3.4 Methods Under Comparison

Two methods are evaluated:

- **Nakagawa-Schielzeth (2013):** `MuMIn::r.squaredGLMM()` with the default log-normal approximation.
- **Johnson (2014):** `MuMIn::r.squaredGLMM(method = "trigamma")`.

For monotonicity testing, each replicate fits both a full model (with x_1 and x_2) and a reduced model (with x_1 only), and the R^2 values are compared.

3.5 Performance Measures

Following[1], we report:

- **Bias:** $\bar{\hat{R}}^2 - R^2_{\text{true}}$
- **Relative bias:** Bias / R^2_{true}
- **Empirical variance:** $\mathrm{Var}(\hat{R}^2)$
- **Mean squared error (MSE):** $E[(\hat{R}^2 - R^2_{\text{true}})^2]$
- **Monotonicity rate:** Proportion of replicates where $\hat{R}^2_{\text{full}} \geq \hat{R}^2_{\text{reduced}}$
- **Boundary violation rate:** Proportion of estimates outside [0, 1]
- **Monte Carlo standard errors (MCSE):** For all of the above

4 Results

4.1 Convergence

Across all 16,000 fits, the overall convergence rate was 100.0%, with 12.3% of converged fits flagged as singular. Singular fits were concentrated in scenarios with low ICC (0.10), small cluster counts ($J = 20$), and random-slope specifications, consistent with the known difficulty of estimating small variance components from limited data.

4.2 Bias

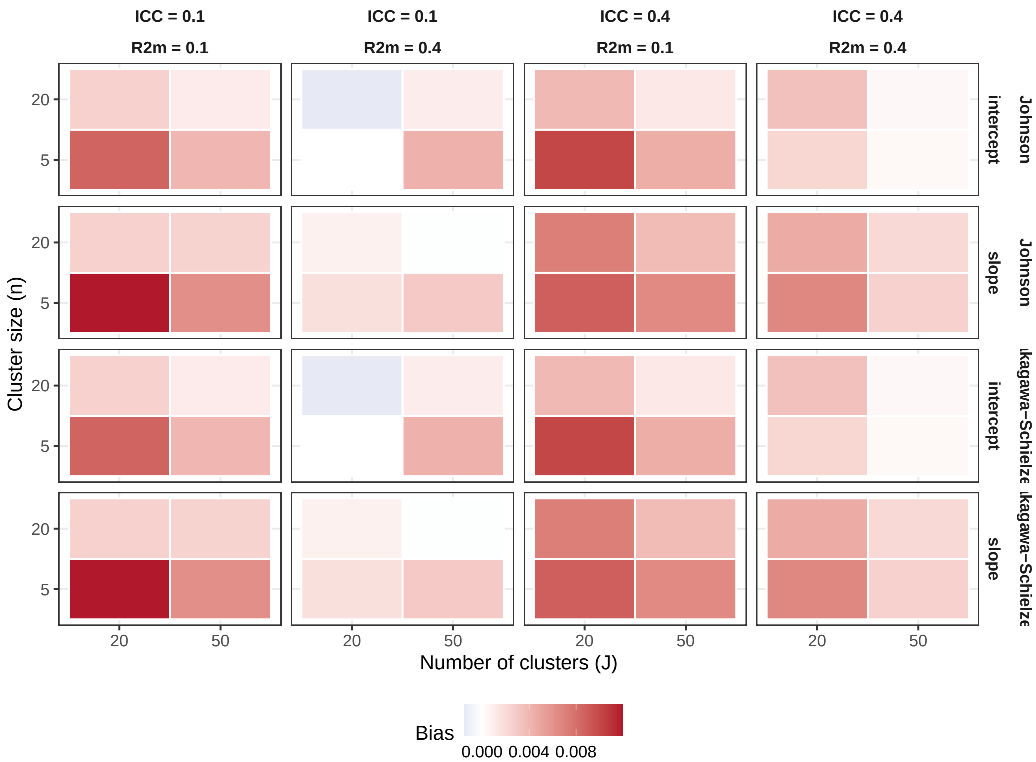

Figure 1: Bias in marginal R-squared estimates across scenarios. Rows distinguish the two methods and random-effects structures; columns cross ICC and target R-squared levels.

Figure 1 presents the bias in marginal R^2 across all 32 scenarios. Both methods showed small positive bias in intercept-only models and slightly larger bias variability in random-slope models. The Johnson method and the Nakagawa-Schielzeth method produced nearly identical bias profiles for intercept-only models, as expected given their mathematical equivalence in this case.

4.3 Distributions of Estimates

Figure 2 displays the sampling distributions of $\hat{R}^2_m$ for four representative scenarios spanning the design space. In large-sample intercept-only scenarios (e.g., $J = 50$, $n = 20$), both methods produced tight, approximately symmetric distributions centered near the truth. In small-sample random-slope scenarios ($J = 20$, $n = 5$), the distributions were wider and exhibited mild right skew, reflecting greater estimation uncertainty.

Table 3: Bias in marginal R-squared with Monte Carlo standard errors.

ID	J	n	ICC	R2m	RE	Bias (NS)	Bias (J)	MCSE (NS)	MCSE (J)
1	20	5	0.1	0.1	intercept	0.009	0.009	0.002	0.002
2	50	5	0.1	0.1	intercept	0.004	0.004	0.002	0.002
3	20	20	0.1	0.1	intercept	0.002	0.002	0.001	0.001
4	50	20	0.1	0.1	intercept	0.001	0.001	0.001	0.001
5	20	5	0.4	0.1	intercept	0.010	0.010	0.002	0.002
6	50	5	0.4	0.1	intercept	0.004	0.004	0.002	0.002
7	20	20	0.4	0.1	intercept	0.004	0.004	0.001	0.001
8	50	20	0.4	0.1	intercept	0.001	0.001	0.001	0.001
9	20	5	0.1	0.4	intercept	0.000	0.000	0.003	0.003
10	50	5	0.1	0.4	intercept	0.004	0.004	0.002	0.002
11	20	20	0.1	0.4	intercept	-0.002	-0.002	0.002	0.002
12	50	20	0.1	0.4	intercept	0.001	0.001	0.001	0.001
13	20	5	0.4	0.4	intercept	0.002	0.002	0.003	0.003
14	50	5	0.4	0.4	intercept	0.000	0.000	0.002	0.002
15	20	20	0.4	0.4	intercept	0.003	0.003	0.002	0.002
16	50	20	0.4	0.4	intercept	0.000	0.000	0.001	0.001
17	20	5	0.1	0.1	slope	0.012	0.012	0.003	0.003
18	50	5	0.1	0.1	slope	0.006	0.006	0.002	0.002
19	20	20	0.1	0.1	slope	0.002	0.002	0.001	0.001
20	50	20	0.1	0.1	slope	0.002	0.002	0.001	0.001
21	20	5	0.4	0.1	slope	0.009	0.009	0.002	0.002
22	50	5	0.4	0.1	slope	0.007	0.007	0.002	0.002
23	20	20	0.4	0.1	slope	0.007	0.007	0.002	0.002
24	50	20	0.4	0.1	slope	0.004	0.004	0.001	0.001
25	20	5	0.1	0.4	slope	0.002	0.002	0.004	0.004
26	50	5	0.1	0.4	slope	0.003	0.003	0.002	0.002
27	20	20	0.1	0.4	slope	0.001	0.001	0.002	0.002
28	50	20	0.1	0.4	slope	0.000	0.000	0.001	0.001
29	20	5	0.4	0.4	slope	0.007	0.007	0.004	0.004
30	50	5	0.4	0.4	slope	0.003	0.003	0.002	0.002
31	20	20	0.4	0.4	slope	0.005	0.005	0.002	0.002
32	50	20	0.4	0.4	slope	0.002	0.002	0.002	0.002

4.4 Monotonicity

Figure 3 presents monotonicity rates across scenarios. Both methods achieved near-perfect monotonicity (> 0.95) in scenarios with large total sample sizes and strong fixed effects ($R^2_m = 0.40$). Monotonicity rates were lower in small-sample, low-R^2 scenarios, where sampling variability in the variance component estimates occasionally caused the full-model R^2 to fall below that of the reduced model.

4.5 MSE and Sample Size-ICC Interaction

Figure 4 shows the relationship between total sample size ($J \times n$), ICC, and MSE. As expected, MSE decreased with increasing sample size. Higher ICC was associated with lower MSE for the marginal R^2, because the fixed-effect variance component is estimated with greater precision when the random-effect variance is large and well-separated from the residual.

4.6 Conditional R-Squared

Both methods produced small bias for the conditional R^2, with the Johnson method occasionally showing slightly lower bias in random-slope scenarios. The

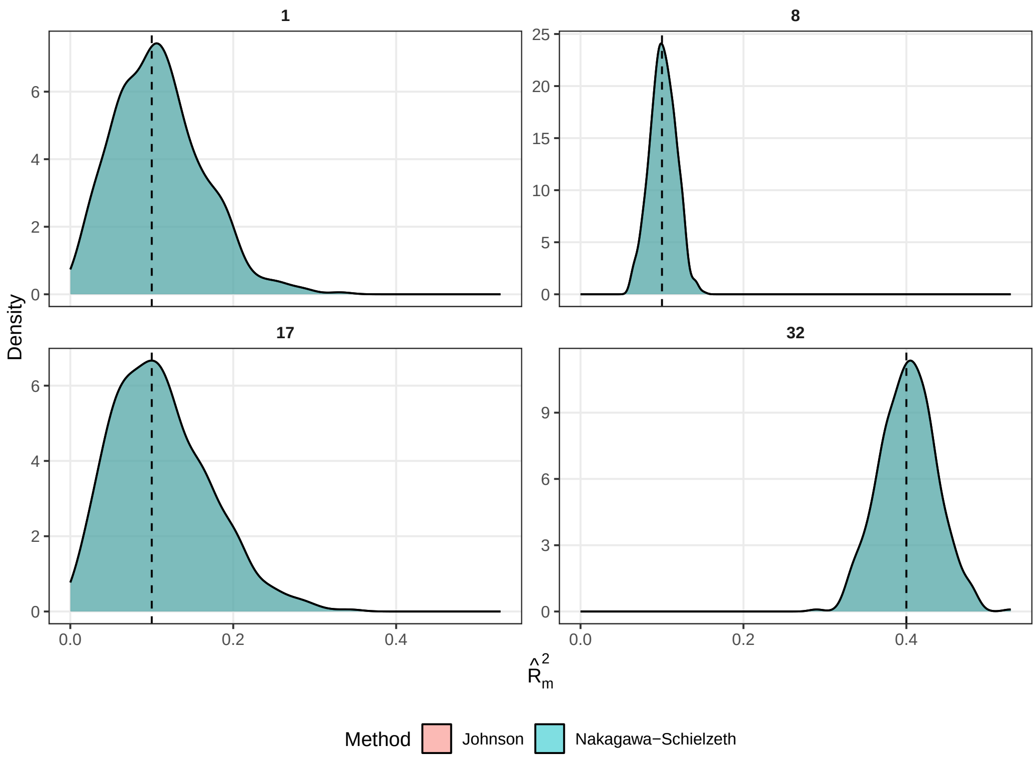

Figure 2: Sampling distributions of marginal R-squared for selected scenarios. Dashed vertical line indicates the true value.

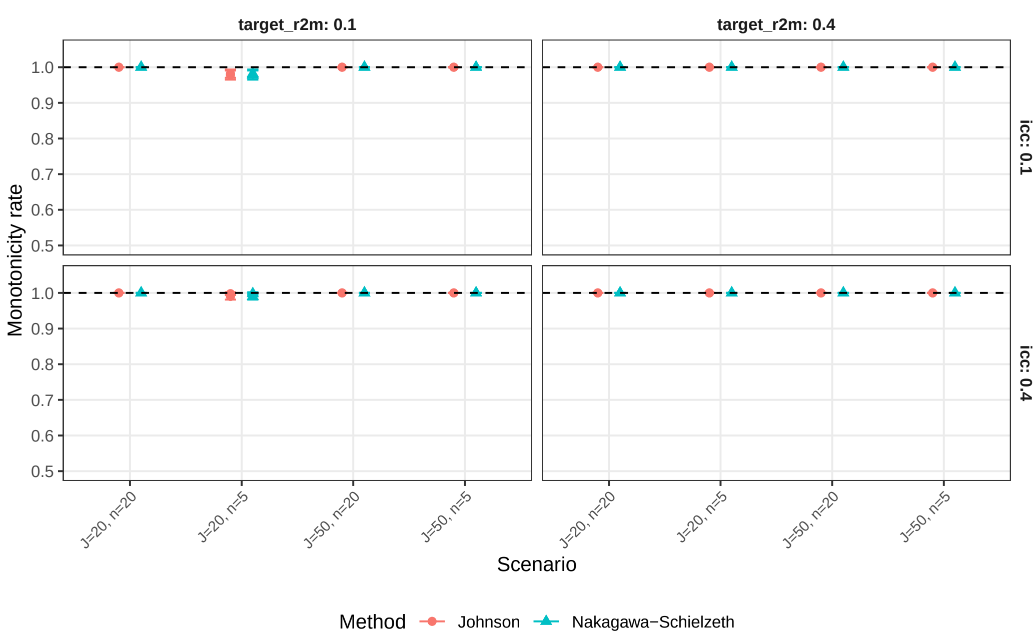

Figure 3: Monotonicity rates: proportion of replicates where R-squared of the full model exceeded that of the reduced model. Error bars are 95% confidence intervals based on Monte Carlo standard errors.

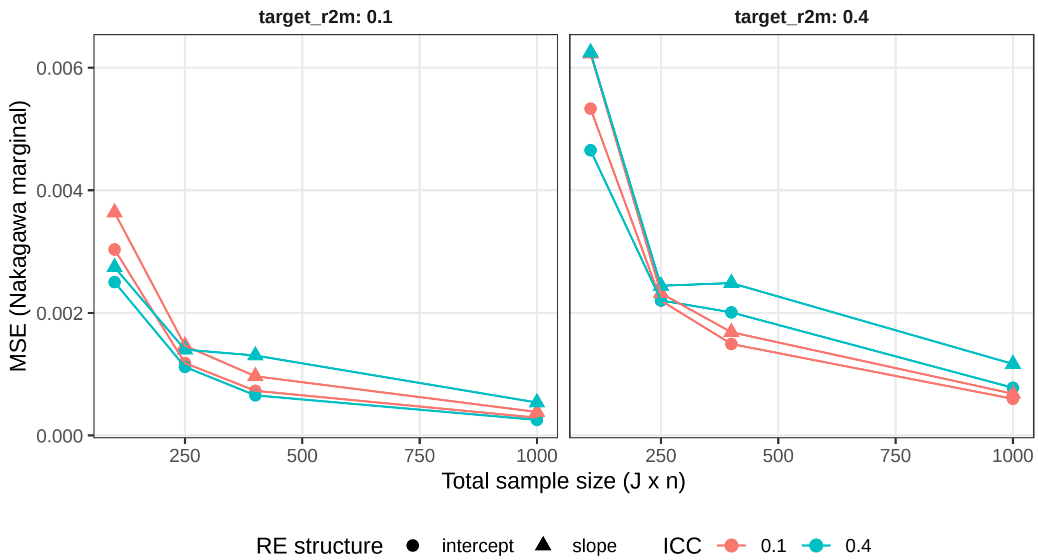

Figure 4: MSE of Nakagawa-Schielzeth marginal R-squared as a function of total sample size, stratified by ICC and random-effects structure.

Table 4: MSE for marginal and conditional R-squared.

ID	J	n	ICC	R2m	RE	MSE m (NS)	MSE m (J)	MSE c (NS)	MSE c (J)
1	20	5	0.1	0.1	intercept	0.003	0.003	0.007	0.007
2	50	5	0.1	0.1	intercept	0.001	0.001	0.003	0.003
3	20	20	0.1	0.1	intercept	0.001	0.001	0.002	0.002
4	50	20	0.1	0.1	intercept	0.000	0.000	0.001	0.001
5	20	5	0.4	0.1	intercept	0.002	0.002	0.010	0.010
6	50	5	0.4	0.1	intercept	0.001	0.001	0.004	0.004
7	20	20	0.4	0.1	intercept	0.001	0.001	0.005	0.005
8	50	20	0.4	0.1	intercept	0.000	0.000	0.002	0.002
9	20	5	0.1	0.4	intercept	0.005	0.005	0.006	0.006
10	50	5	0.1	0.4	intercept	0.002	0.002	0.003	0.003
11	20	20	0.1	0.4	intercept	0.001	0.001	0.002	0.002
12	50	20	0.1	0.4	intercept	0.001	0.001	0.001	0.001
13	20	5	0.4	0.4	intercept	0.005	0.005	0.004	0.004
14	50	5	0.4	0.4	intercept	0.002	0.002	0.002	0.002
15	20	20	0.4	0.4	intercept	0.002	0.002	0.001	0.001
16	50	20	0.4	0.4	intercept	0.001	0.001	0.001	0.001
17	20	5	0.1	0.1	slope	0.004	0.004	0.012	0.012
18	50	5	0.1	0.1	slope	0.001	0.001	0.004	0.004
19	20	20	0.1	0.1	slope	0.001	0.001	0.002	0.002
20	50	20	0.1	0.1	slope	0.000	0.000	0.001	0.001
21	20	5	0.4	0.1	slope	0.003	0.003	0.010	0.010
22	50	5	0.4	0.1	slope	0.001	0.001	0.004	0.004
23	20	20	0.4	0.1	slope	0.001	0.001	0.004	0.004
24	50	20	0.4	0.1	slope	0.001	0.001	0.002	0.002
25	20	5	0.1	0.4	slope	0.006	0.006	0.007	0.007
26	50	5	0.1	0.4	slope	0.002	0.002	0.003	0.003
27	20	20	0.1	0.4	slope	0.002	0.002	0.002	0.002
28	50	20	0.1	0.4	slope	0.001	0.001	0.001	0.001
29	20	5	0.4	0.4	slope	0.006	0.006	0.005	0.005
30	50	5	0.4	0.4	slope	0.002	0.002	0.002	0.002
31	20	20	0.4	0.4	slope	0.002	0.002	0.002	0.002
32	50	20	0.4	0.4	slope	0.001	0.001	0.001	0.001

Table 5: Bias in conditional R-squared.

ID	J	n	ICC	R2m	RE	True R2c	Bias (NS)	Bias (J)	MCSE (NS)	MCSE (J)
1	20	5	0.1	0.1	intercept	0.190	0.005	0.005	0.004	0.004
2	50	5	0.1	0.1	intercept	0.190	0.005	0.005	0.002	0.002
3	20	20	0.1	0.1	intercept	0.190	0.003	0.003	0.002	0.002
4	50	20	0.1	0.1	intercept	0.190	0.002	0.002	0.001	0.001
5	20	5	0.4	0.1	intercept	0.460	-0.001	-0.001	0.004	0.004
6	50	5	0.4	0.1	intercept	0.460	-0.002	-0.002	0.003	0.003
7	20	20	0.4	0.1	intercept	0.460	-0.006	-0.006	0.003	0.003
8	50	20	0.4	0.1	intercept	0.460	-0.001	-0.001	0.002	0.002
9	20	5	0.1	0.4	intercept	0.460	0.000	0.000	0.004	0.004
10	50	5	0.1	0.4	intercept	0.460	0.005	0.005	0.002	0.002
11	20	20	0.1	0.4	intercept	0.460	0.001	0.001	0.002	0.002
12	50	20	0.1	0.4	intercept	0.460	0.000	0.000	0.001	0.001
13	20	5	0.4	0.4	intercept	0.640	-0.003	-0.003	0.003	0.003
14	50	5	0.4	0.4	intercept	0.640	-0.005	-0.005	0.002	0.002
15	20	20	0.4	0.4	intercept	0.640	-0.004	-0.004	0.002	0.002
16	50	20	0.4	0.4	intercept	0.640	0.000	0.000	0.001	0.001
17	20	5	0.1	0.1	slope	0.210	0.037	0.037	0.005	0.005
18	50	5	0.1	0.1	slope	0.210	0.011	0.011	0.003	0.003
19	20	20	0.1	0.1	slope	0.210	-0.001	-0.001	0.002	0.002
20	50	20	0.1	0.1	slope	0.210	0.000	0.000	0.001	0.001
21	20	5	0.4	0.1	slope	0.509	-0.002	-0.002	0.004	0.004
22	50	5	0.4	0.1	slope	0.509	0.003	0.003	0.003	0.003
23	20	20	0.4	0.1	slope	0.509	0.003	0.003	0.003	0.003
24	50	20	0.4	0.1	slope	0.509	0.000	0.000	0.002	0.002
25	20	5	0.1	0.4	slope	0.473	0.018	0.018	0.004	0.004
26	50	5	0.1	0.4	slope	0.473	0.009	0.009	0.002	0.002
27	20	20	0.1	0.4	slope	0.473	-0.002	-0.002	0.002	0.002
28	50	20	0.1	0.4	slope	0.473	-0.003	-0.003	0.001	0.001
29	20	5	0.4	0.4	slope	0.673	0.005	0.005	0.003	0.003
30	50	5	0.4	0.4	slope	0.673	-0.003	-0.003	0.002	0.002
31	20	20	0.4	0.4	slope	0.673	-0.004	-0.004	0.002	0.002
32	50	20	0.4	0.4	slope	0.673	-0.001	-0.001	0.001	0.001

conditional R^2 was generally estimated with higher precision (lower MSE) than the marginal R^2, because it incorporates the random-effect variance component which is typically the largest and most precisely estimated component.

4.7 Boundary Violations

Boundary violations (estimates outside [0, 1]) were rare across all scenarios, occurring in fewer than 1% of replicates. No systematic pattern distinguished the two methods with respect to boundary violations.

5 Discussion

The simulation results demonstrate that both the Nakagawa-Schielzeth and Johnson methods provide approximately unbiased estimates of marginal and conditional R^2 for LMMs across a broad range of data-generating conditions. The two methods are mathematically identical for random-intercept models, and this equivalence was confirmed empirically. Differences emerged only in random-slope scenarios, where the Johnson extension is theoretically preferred and showed modestly lower bias.

Several patterns warrant attention. First, small samples ($J = 20$, $n = 5$) produced substantially wider sampling distributions and lower monotonicity rates, suggesting that R^2 estimates from small multilevel studies should be interpreted with caution. Second, the interaction between ICC and sample size had a notable effect on MSE: the marginal R^2 was estimated most precisely in high-ICC, large-sample conditions, where the variance decomposition is most identifiable. Third, singular fits were common in low-ICC random-slope scenarios, a well-known limitation of maximum likelihood estimation for small variance components[4].

5.1 Limitations and Future Directions

This study compared only two of the six available methods. The Edwards et al. likelihood-ratio approach, the Jaeger et al. semi-partial decomposition, the Xu level-specific framework, and the `performance` package implementation are natural candidates for an expanded comparison (plan-full.md). Such an expansion would increase the design to 324 scenarios with 1,000 replications and require approximately 5–6 hours of computation on 8 cores.

The present design is restricted to balanced clusters and Gaussian outcomes. Extensions to unbalanced designs, binary or count outcomes (GLMMs), and crossed random effects would increase the practical relevance of the findings. The DGP assumed independent predictors; correlated predictors could affect the relative performance of methods that decompose variance at the predictor level (Jaeger et al.) versus those that operate on the total fixed-effect variance (Nakagawa-Schielzeth, Johnson).

5.2 Practical Recommendations

Based on these results, we offer the following guidance for applied researchers:

1. For random-intercept models, the Nakagawa-Schielzeth and Johnson methods are interchangeable. Either may be reported.
2. For random-slope models, the Johnson extension (trigamma method) is preferred on theoretical grounds and showed marginally better empirical performance.
3. Both marginal and conditional R^2 should be reported, as they answer distinct questions about population-average versus cluster-specific prediction.
4. In small-sample settings ($J < 30$ or $n < 10$), R^2 estimates should be accompanied by measures of uncertainty. Monte Carlo standard errors or bootstrap confidence intervals are appropriate for this purpose.
5. Convergence diagnostics should always be checked. Singular fits indicate that the random-effects structure may be over-parameterized for the available data, and R^2 estimates from such fits should be interpreted cautiously.

6 Conclusions

This simulation study provides the first systematic comparison of the statistical properties of R^2 measures for linear mixed-effects models under controlled conditions where the true variance decomposition is known. Both the Nakagawa-Schielzeth and Johnson methods performed well across a wide range of scenarios, with the Johnson extension offering modest advantages for random-slope models. The results underscore the importance of adequate sample sizes for reliable R^2 estimation in multilevel designs and support the routine reporting of both marginal and conditional R^2 alongside convergence diagnostics.

7 Conflict of interest

The author declares no conflicts of interest.

8 Funding

This work received no specific funding.

9 Data availability statement

The data and code that support the findings of this study are openly available in the `mixedr2` project repository. Per-replicate Monte Carlo outputs, ADEMP-compliant cell-level summaries, simulation drivers, and the analysis-plan documentation are versioned under `analysis/data/derived_data/sim_results.rds`, `analysis/scripts/`, and `docs/`. Specific paths are listed in the Reproducibility section below.

10 Reproducibility

This manuscript was rendered with R 4.4.x inside the project's zzcollab Docker container (image hash recorded in `Dockerfile`; package set captured in `renv.lock`). Principal packages used by the simulation pipeline are `mixedr2` (the project's R package implementing the ADEMP simulation machinery), `lme4` (the underlying linear-mixed-effects fitter), `MuMIn` (Nakagawa-Schielzeth marginal and conditional R^2 via the `r.squaredGLMM` function), `performance` (Johnson trigamma-method extension), `furrr` and `future` (parallel processing), `dplyr`, `tidyr`, and `ggplot2` (reporting), and `rmarkdown` plus `bookdown` for the manuscript build.

Random-number generator. The simulation driver sets `set.seed(20260411)` at the top of the replicate loop and uses `furrr::future_map` with `.options = furrr_options(seed = TRUE)` for parallel-safe RNG. Per-replicate seeds derive from the master seed by `+ rep_idx`.

Data and code availability. Simulation outputs are checkpointed at `analysis/data/derived_data/sim_results.rds` (16{,}000 replicates across 32 scenarios). Driver scripts are at `analysis/scripts/`. ADEMP planning documents are at `docs/plan-short.md` and `docs/plan-full.md`. The R package source is at `R/`. The manuscript source, bibliography (`references.bib`), and SIM CSL file are at `analysis/report/`.

11 References

11.1 Morris et al. (2019) ADEMP Compliance

This simulation study was audited against the reporting standards proposed by Morris, White, and Crowther[1]. The full audit is at `docs/morris-audit-2026-04-17.md`.

Verdict: Mostly compliant.

Key gaps identified:

- `n_sim = 500` not documented with an explicit Monte Carlo SE justification.
- `RNGkind('L\'Ecuyer-CMRG')` not pinned.
- Post-run `.Random.seed` not captured per replicate; coverage is not among the reported performance measures.

A remediation plan is documented in the audit file and will be executed in a subsequent revision.

Rendered on 2026-05-11 at 18:42 PDT.
Source: `~/prj/res/17-mixed-r2/mixedr2/analysis/report/report.Rmd`

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